PRACTICAL 2
CREATE AND CONFIGURE VM INSTANCES ON GOOGLE CLOUD PLATFORM

Practical objective:
In this practical, you will create, configure, connect to, inspect, stop, start, and delete a virtual machine instance on Google Cloud Platform using Compute Engine.

Cloud platform used:
Google Cloud Platform, also called GCP.

Main service used:
Compute Engine.

Estimated outcome:
By the end of this practical, you will be able to:
- Enable or open Compute Engine in Google Cloud Console.
- Create a VM instance with basic configuration.
- Understand machine type, region, zone, boot disk, firewall, labels, and service account options.
- Connect to a VM using SSH.
- Run basic Linux commands inside a VM.
- Stop and start a VM.
- Delete a VM to avoid unnecessary cost.
- Review the VM configuration using current Google Cloud best practices.

Important safety message:
Virtual machines can create cloud charges while they are running, and disks can also create charges even when a VM is stopped. Delete test resources after the practical unless your teacher instructs you to keep them.

1. BASIC THEORY BEFORE STARTING

1.1 What is a virtual machine?

A virtual machine, or VM, is a software-based computer that runs on physical hardware managed by a cloud provider.

A VM has:
- CPU
- Memory
- Operating system
- Disk
- Network interface
- IP address

1.2 What is Google Compute Engine?

Google Compute Engine is the Google Cloud service used to create and manage virtual machines.

Compute Engine provides:
- VM instances
- Machine types
- Persistent disks
- Images
- Snapshots

- Instance groups
- Load balancing support
- Firewall integration

Compute Engine is an example of Infrastructure as a Service because it gives you control over virtual machines, operating systems, disks, and networking.

1.3 What is a machine type?

A machine type defines the amount of virtual CPU and memory assigned to a VM.

Examples:
- e2-micro
- e2-small
- e2-medium
- n2-standard-2

Choose a small machine type for lab work. Bigger machines cost more and are usually not needed for basic practicals.

1.4 What is a region and zone?

A region is a geographic location.

Example:
- us-central1
- asia-south1
- europe-west1

A zone is a location inside a region.

Example:
- us-central1-a
- us-central1-b
- asia-south1-a

Regions and zones decide where your VM physically runs. Choose a nearby region when possible to reduce latency, but follow your teacher's instructions for lab consistency.

1.5 What is a boot disk?

A boot disk contains the operating system and files required to start the VM.

Examples of operating system images:
- Debian
- Ubuntu
- CentOS Stream
- Rocky Linux
- Windows Server

The boot disk is like the hard disk of a computer where the operating system is installed.

1.6 What is a firewall rule?

A firewall rule controls network traffic allowed to or from a VM.

Examples:

- Allow SSH on port 22.
- Allow HTTP on port 80.
- Allow HTTPS on port 443.

Firewall rules are security gates. Do not open ports to the whole internet unless you understand why it is needed.

1.7 What is SSH?

SSH stands for Secure Shell. It is used to connect securely to a Linux VM and run commands remotely.

SSH lets you operate a cloud VM from your browser or terminal, similar to using a keyboard and screen attached to a remote computer.

1.8 What is OS Login?

OS Login is a Google Cloud feature that manages Linux VM SSH access using IAM permissions instead of manually managing SSH keys in metadata.

OS Login is a modern and recommended way to manage SSH access because access can be controlled through IAM and revoked when a user no longer has permission.

1.9 What is Shielded VM?

Shielded VM provides security features that help protect VMs from rootkits and boot-level malware.

Shielded VM features include:
- Secure Boot
- Virtual Trusted Platform Module, or vTPM
- Integrity monitoring

Shielded VM is like extra startup protection for your virtual machine. Keep these features enabled unless there is a special reason to change them.

2. REQUIREMENTS

2.1 Hardware requirements

- Computer or laptop
- Internet connection

2.2 Software requirements

- Web browser

2.3 Cloud requirements

- Google Cloud account
- Google Cloud project
- Billing enabled if required for Compute Engine
- Permission to create VM instances

Required IAM permission concept:
To create VM instances, a user commonly needs a role such as Compute Instance Admin or suitable custom permissions assigned by the administrator.

If you cannot create a VM, it may be because billing is disabled or your account does not have enough permissions.

3. STEP-BY-STEP PRACTICAL

STEP 1: Open Google Cloud Console

Open:

https://console.cloud.google.com

Sign in with your Google Cloud account.

Use the same account and project assigned for your practical work. Using a wrong account may hide your project or create resources in the wrong place.

Expected result:
Google Cloud Console opens successfully.

STEP 2: Select the correct project

Use the project selector at the top of the console and choose the project for this practical.

Every VM belongs to a project. Always confirm the project before creating cloud resources.

Expected result:
The correct project name is visible in the top navigation bar.

STEP 3: Open Compute Engine

In the search bar, type:

Compute Engine

Open Compute Engine.

Compute Engine is the service where VM instances are created and managed.

Expected result:
The Compute Engine page opens.

STEP 4: Enable Compute Engine API if asked

If Google Cloud asks you to enable the Compute Engine API, click:

Enable

Wait for the service to initialize.

Google Cloud services use APIs. A service must be enabled in the project before you can create resources from that service.

Expected result:

Compute Engine becomes available for the project.

STEP 5: Open VM instances page

In Compute Engine, click:

VM instances

The VM instances page lists all virtual machines created in the selected project.

Expected result:
The VM instances list page opens.

STEP 6: Click Create Instance

Click:

Create Instance

This starts the VM creation form. You will select the VM name, location, machine type, operating system, disk, network, and security options.

Expected result:
The Create an instance page opens.

STEP 7: Enter VM name

Enter a VM name such as:

cloudnative-vm-1

Use lowercase letters, numbers, and hyphens.

Use a meaningful name so you can identify the VM later. Avoid spaces and special characters.

Expected result:
The VM name is entered.

STEP 8: Select region and zone

Choose a region and zone as instructed by your teacher.

Example:

Region: us-central1
Zone: us-central1-a

For lab work, any available region may work, but choosing the same region for all the class makes troubleshooting easier.

Expected result:
Region and zone are selected.

STEP 9: Select machine type

Choose a small machine type for lab work.

Example:

Series: E2
Machine type: e2-micro or e2-small

Small machine types are enough for learning Linux commands and basic web server tests. Larger machine types cost more.

Expected result:
A suitable small machine type is selected.

STEP 10: Configure boot disk

Click the boot disk option and choose a Linux operating system.

Recommended beginner option:

Operating system: Debian
Version: Latest stable Debian version shown in the console
Boot disk type: Balanced persistent disk or default option
Size: 10 GB or default lab size

Debian is a common Linux distribution and works well for beginner cloud VM practicals. Use default disk settings unless your teacher asks otherwise.

Expected result:
Boot disk is configured.

STEP 11: Review identity and API access

Find the section related to service account or identity.

For a basic lab, keep the default setting if your teacher allows it, or choose a least-privilege service account if provided.

A VM service account controls what Google Cloud APIs the VM can access. In production, avoid using overly powerful service accounts.

Best-practice note:
Use least privilege. Do not attach a highly privileged service account to a VM unless required.

Expected result:
Service account setting is reviewed.

STEP 12: Add labels

Find the labels section and add labels such as:

Key: purpose
Value: practical

Key: owner
Value: student

Labels help organize resources, filter them, and understand cost or ownership. Using labels is a good cloud management habit.

Expected result:
Labels are added to the VM configuration.

STEP 13: Review firewall options

In the firewall section, you may see options such as:
- Allow HTTP traffic
- Allow HTTPS traffic

For this basic VM practical, leave them unchecked unless your teacher asks you to install a web server.

Opening HTTP or HTTPS allows external traffic to reach your VM. Do not open ports unless the practical requires it.

Expected result:
Firewall options are reviewed.

STEP 14: Review advanced security options

Look for security settings such as Shielded VM.

Keep Shielded VM protections enabled if they are available by default.

Shielded VM features provide extra protection during VM boot. Keeping them enabled follows modern cloud security practice.

Expected result:
Security options are reviewed.

STEP 15: Create the VM

Review all settings, then click:

Create

After clicking Create, Google Cloud provisions the VM. Provisioning means cloud resources such as CPU, memory, disk, and network are assigned to your instance.

Expected result:
The VM appears in the VM instances list.

STEP 16: Wait until VM is running

Wait until the VM status becomes:

Running

A VM needs some time to start, just like a physical computer needs time to boot.

Expected result:
The VM is running.

STEP 17: Observe VM details

In the VM instances list, observe:
- Name
- Zone
- Internal IP
- External IP, if assigned
- Status

Internal IP is used inside the cloud network. External IP is used to reach the VM from the internet. For stronger security, production VMs often avoid public external IPs and use controlled access methods.

Expected result:
Basic VM details are identified.

STEP 18: Connect to the VM using SSH

Click the SSH button next to the VM.

Wait for the browser-based SSH window to open.

Browser SSH is convenient for learning. In secure production environments, organizations commonly use OS Login and restricted network access such as Identity-Aware Proxy.

Expected result:
An SSH terminal opens.

STEP 19: Run hostname command

Inside the SSH terminal, run:

```
hostname
```

The hostname command shows the name of the machine you are connected to. It confirms that your SSH session is inside the VM.

Expected result:
The VM hostname appears.

STEP 20: Check operating system information

Run:

```
cat /etc/os-release
```

Expected result:
Operating system details are displayed.

STEP 21: Check disk usage

Run:

df -h

Expected result:
Disk usage information appears.

STEP 22: Check memory usage

Run:

free -h

Expected result:
Memory usage information appears.

STEP 23: Check CPU information

Run:

lscpu

Expected result:
CPU information appears.

STEP 24: Update package information

For Debian or Ubuntu, run:

sudo apt update

Expected result:
Package information is updated.

STEP 25: Optional web server test

Only if your teacher asks you to test HTTP traffic, install Nginx:

sudo apt install -y nginx

Then check status:

systemctl status nginx

Nginx is a web server. If HTTP firewall traffic is not allowed, the web server may run inside the VM but not be reachable from the public internet.

Expected result:

Nginx is installed and running if this optional step is performed.

STEP 26: Exit SSH session

Run:

exit

The exit command closes the SSH session and returns you to the console.

Expected result:
SSH session is closed.

STEP 27: Stop the VM

In the VM instances page, select the VM and click:

Stop

Confirm if asked.

Stopping a VM stops CPU and memory charges, but persistent disk and reserved IP charges may still apply. Delete resources if the practical is complete.

Expected result:
VM status changes to stopped or terminated.

STEP 28: Start the VM again

Select the stopped VM and click:

Start / Resume

Starting a VM boots it again. A stopped VM can be restarted later if the disk is kept.

Expected result:
VM returns to running status.

STEP 29: Delete the VM after practical

After completing observations, select the VM and click:

Delete

Confirm deletion.

Deleting test VMs prevents unnecessary charges. Always clean up resources after cloud practicals unless your teacher asks you to keep them.

Expected result:
The VM is deleted from the VM instances list.

STEP 30: Check that no test VM remains

Refresh the VM instances page and confirm that the test VM is no longer listed.

Cloud cleanup is an important professional habit. Many real cloud bills increase because unused resources are forgotten.

Expected result:
No unnecessary VM instance remains.

4. OPTIONAL COMMAND-LINE METHOD

This section is optional and should be used only if your teacher wants you to practice the Google Cloud CLI.

4.1 Open Cloud Shell

Open Cloud Shell from the Google Cloud Console.

Cloud Shell includes the gcloud command-line tool, which can create and manage Google Cloud resources.

4.2 Check active project

Run:

```
gcloud config get-value project
```

Always confirm the active project before running cloud commands.

4.3 Create a VM using gcloud

Example command:

```
gcloud compute instances create cloudnative-vm-cli \
  --zone=us-central1-a \
  --machine-type=e2-micro \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --labels=purpose=practical,owner=student
```

4.4 Delete the CLI-created VM

Run:

```
gcloud compute instances delete cloudnative-vm-cli --zone=us-central1-a
```

Always delete command-line test resources after the practical.

5. OBSERVATION TABLE

Write your observations in this format:

VM Name:

Project ID:

Region:

Zone:

Machine type:

Operating system:

Boot disk size:

Internal IP:

External IP, if any:

Firewall options selected:

Labels added:

Commands executed:
- hostname
- cat /etc/os-release
- df -h
- free -h
- lscpu

Was the VM deleted after practical?

6. RESULT

A VM instance was created and configured using Google Compute Engine. Basic VM details were observed, SSH access was tested, Linux commands were executed, and the VM was stopped, started, and deleted after completion.

7. CONCLUSION

This practical covered how to create and configure a Google Cloud Compute Engine VM instance. They also learned the importance of selecting the correct project, choosing a suitable machine type, reviewing security settings, using SSH, observing system information, and deleting unused resources to avoid unnecessary cost.

8. LATEST-PRACTICES REVIEW TEST

This review checks whether the practical follows current recommended Google Cloud VM practices.

1. Was the correct project selected before creating the VM?
   Yes. Creating resources in the wrong project causes confusion, permission problems, and billing mistakes.

2. Was the Compute Engine API enabled only for the required project?
   Yes. Google Cloud services must be enabled at the project level before use.

3. Was a small, right-sized machine type used for lab work?

Yes. Lab VMs should use small machine types such as E2 small options unless more resources are required.

4. Were labels added?
Yes. Labels help with organization, ownership, filtering, and cost tracking.

5. Were unnecessary firewall ports avoided?
Yes. HTTP or HTTPS should be opened only when the practical requires web access.

6. Was SSH access treated carefully?
Yes. For learning, browser SSH is acceptable. For stronger practice, use OS Login
and restrict SSH network access, preferably through Identity-Aware Proxy where available.

7. Were Shielded VM protections kept enabled?
Yes. Shielded VM features help protect boot integrity and should normally remain enabled.

8. Was least privilege considered for the VM service account?
Yes. Avoid attaching highly privileged service accounts to VMs.

9. Was the VM deleted after the practical?
Yes. Deleting test resources avoids unnecessary charges.

10. Would production workloads use a single manually created VM?
Usually no. Production workloads often use automation, instance templates, managed instance groups, autoscaling, load balancing, monitoring, and stronger security controls.

If the points above are covered, the practical aligns with current platform practices.
Compute Engine practices and introduces current operational recommendations.

9. VIVA QUESTIONS

1. What is a virtual machine?
2. What is Google Compute Engine?
3. What is a machine type?
4. What is the difference between region and zone?
5. What is a boot disk?
6. What is a firewall rule?
7. What is SSH?
8. What is OS Login?
9. Why should Shielded VM be enabled?
10. Why should labels be added to cloud resources?
11. Why should unused VMs be deleted?
12. What is the difference between internal IP and external IP?

10. SHORT ANSWERS FOR VIVA

1. A virtual machine is a software-based computer running on cloud infrastructure.
2. Google Compute Engine is the GCP service used to create and manage VMs.
3. A machine type defines CPU and memory assigned to a VM.
4. A region is a geographic area, and a zone is a location inside a region.
5. A boot disk contains the operating system used to start the VM.
6. A firewall rule controls allowed or denied network traffic.
7. SSH is a secure method to connect to a Linux VM remotely.
8. OS Login manages SSH access using IAM permissions.
9. Shielded VM helps protect VM boot integrity and security.
10. Labels help organize resources and track ownership or cost.
11. Unused VMs should be deleted to avoid unnecessary cloud charges.
12. Internal IP is used inside the cloud network; external IP is used from the internet.

11. COMMON MISTAKES TO AVOID

1. Creating the VM in the wrong project.
2. Choosing a large machine type for a simple lab.
3. Opening HTTP or HTTPS firewall rules without need.
4. Forgetting to delete the VM after practical.
5. Ignoring persistent disk charges.
6. Attaching a highly privileged service account.
7. Sharing SSH access or credentials.
8. Not checking region and zone.
9. Not adding labels.
10. Treating browser SSH as the only secure production method.

12. PRACTICAL COMPLETION CHECKLIST

Before submitting this practical, confirm:

1. You opened Google Cloud Console.
2. You selected the correct project.
3. You opened Compute Engine.
4. You enabled Compute Engine API if required.
5. You opened VM instances page.
6. You created a VM with a proper name.
7. You selected region and zone.
8. You selected a small machine type.
9. You selected a Linux boot disk.
10. You reviewed service account settings.
11. You added labels.
12. You reviewed firewall options.
13. You kept security options enabled where possible.
14. You connected using SSH.
15. You ran basic Linux commands.
16. You observed VM details.
17. You stopped and started the VM.
18. You deleted the VM after practical.
19. You completed the latest-practices review test.
20. You answered the viva questions.